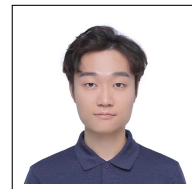


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Education

- 2022–Present **Ph.D. in Computer Science**, *Vanderbilt University*, Nashville, TN, US
Advisor: Prof. Abhishek Dubey
- 2018–2021 **B.S. in Computer Science**, *Vanderbilt University*, Nashville, TN, US
Double Major: Computer Science, Mathematics

Research Interests

- **Sequential Decision-Making Under Partial Observability:** Advanced algorithms for POMDPs including particle-based belief representation, annealed importance sampling, and Monte Carlo tree search with applications to real-time planning under uncertainty.
- **Adaptive Planning in Non-Stationary Environments:** Online learning and planning methods that adapt to changing environmental dynamics, including in-context learning with temporal abstractions and policy-augmented search for evolving systems.
- **AI for Time-Critical Applications:** Deployment of decision-theoretic algorithms to safety-critical domains including UAV search and rescue operations, smart grid vehicle-to-building systems, and emergency response coordination.

Work Experience

- May–Aug 2026 **Siemens Technology**, *Research Intern, Cybersecurity & AI*, Princeton, NJ
- Contributed to developing a simulation environment for training autonomous defense agents, modeled after real-world power grid and water treatment systems, with policies successfully transferred and evaluated on physical infrastructure.
 - Developed CPOMCP, a POMDP-based online planning agent that combines Monte Carlo tree search with causal action pruning to eliminate irrelevant actions before search, reducing the planning space and achieving strong defense performance against adaptive attackers in a partially observable multi-agent setting.
 - Implemented DreamerV3 from scratch in PyTorch — including a recurrent state-space model (RSSM), actor-critic trained entirely in latent-space imagination, and a learned action mask predictor — and used it as the planning backbone to replace environment rollouts with world-model imagination, significantly speeding up planning while maintaining defense performance.
 - Developed an autonomous defense agent powered by a locally-hosted Qwen2.5-coder LLM that performs zero-shot threat analysis on live network telemetry, reasons across multiple network zones, and autonomously executes mitigation actions.
- 2023–Present **DARPA ANSR Program**, *Graduate Research Assistant*, Vanderbilt University
- Developed "Shrinking POMCP" algorithm that reduces computational complexity by dynamically guiding UAV search toward non-sparse belief regions, achieving significant improvements over standard MCTS in time-constrained urban search and rescue scenarios. Integrated AirSim-ROS2 simulator with probabilistic world model and neuro-symbolic navigator for real-time path planning under partial observability, enabling safe navigation in complex 3D environments [\[paper\]](#)
 - Applied neuro-symbolic AI techniques combining neural perception with symbolic reasoning for UAV search and rescue, contributing to *Neuro-symbolic AI: Foundations and Applications* (Wiley-IEEE Press, 2025) [\[book\]](#)
 - Extending framework to multi-agent pursuit-evasion problems in urban environments with dynamic obstacles and adversarial targets (ongoing)

- 2023–2024 **Nissan Motor Corporation, Research Collaborator**
- Designed DG-MCTS algorithm for vehicle-to-building (V2B) energy optimization using real-world EV data [paper]
 - Optimized charging schedules reducing peak demand charges while meeting user requirements [paper]
 - Deployed solution for heterogeneous charger configurations and time-of-use pricing [paper]
- 2022–Present **Belief Representation for Partial Observability, Graduate Researcher, Vanderbilt University**
- Developed ESCORT algorithm using Stein-variational methods with correlation-aware projections for multi-modal belief distributions [paper] [code]
 - Created AIROAS framework combining annealed importance resampling with observation adaptation for POMDPs [paper]
 - Advanced in-context planning (I-TAP) using latent temporal abstractions for high-dimensional continuous control (*under review, TMLR*) [arXiv]
- 2022–Present **Adaptive Planning for Non-Stationary Systems, Graduate Researcher, Vanderbilt University**
- Created NS-Gym, first standardized toolkit for non-stationary MDPs with comprehensive benchmark environments [paper] [code]
 - Developed ADA-MCTS algorithm that balances pessimistic and optimistic planning based on epistemic and aleatoric uncertainty in changing environments [paper] [code]
 - Designed PA-MCTS that combines stale policies with online search to maintain performance when environmental dynamics change [paper] [code]

Publications

My research advances sequential decision-making under uncertainty through both theoretical contributions and practical deployments. I develop algorithms for fundamental challenges in partial observability and non-stationary environments, demonstrating their effectiveness in time-critical applications from autonomous search and rescue to smart grid optimization, while creating open-source benchmarks and toolkits that enable reproducible research in the community:

Algorithmic Innovations in Sequential Decision-Making

- [1] Yunuo Zhang, Baiting Luo, Ayan Mukhopadhyay, Gabor Karsai, and Abhishek Dubey. “ESCORT: Efficient Stein-variational and Sliced Consistency-Optimized Temporal Belief Representation for POMDPs”. In: *Proceedings of the Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS)*. 2025. URL: <https://neurips.cc/virtual/2025/poster/115751>.
 ■ Summary: Develops particle-based belief representation using Stein-variational methods with correlation-aware projections to accurately model multi-modal distributions in POMDPs.
- [2] Yunuo Zhang, Baiting Luo, Ayan Mukhopadhyay, and Abhishek Dubey. “Observation Adaptation via Annealed Importance Resampling for Partially Observable Markov Decision Processes”. In: *Proceedings of the 35th International Conference on Automated Planning and Scheduling (ICAPS)*. accepted as oral presentation. AAAI Press, 2025. URL: <https://ojs.aaai.org/index.php/ICAPS/article/view/36132/38286>.
 ■ Summary: Proposes annealed importance resampling to address particle degeneracy in online POMDP solvers when observations are highly informative, improving belief approximation accuracy.
- [3] Baiting Luo, Yunuo Zhang, Nathaniel S. Keplinger, Samir Gupta, Abhishek Dubey, and Ayan Mukhopadhyay. *In-Context Planning with Latent Temporal Abstractions*. Under review at Transactions on Machine Learning Research (TMLR). 2026. arXiv: [2602.18694](https://arxiv.org/abs/2602.18694) [cs.LG]. URL: <https://arxiv.org/abs/2602.18694>.
 ■ Summary: Unifies in-context adaptation and online planning in learned latent temporal-abstraction space, enabling efficient decision-making under stochastic dynamics and partial observability.
- [4] Baiting Luo, Yunuo Zhang, Abhishek Dubey, and Ayan Mukhopadhyay. “Act as You Learn: Adaptive Decision-Making in Non-Stationary Markov Decision Processes”. In: *Proceedings of the 23rd International Conference on Autonomous Agents and Multiagent Systems*. AAMAS '24. Auckland, New Zealand: International Foundation for Autonomous Agents and Multiagent Systems, 2024, pp. 1301–1309. ISBN: 9798400704864.
 ■ Summary: Introduces ADA-MCTS algorithm that adaptively balances pessimistic and optimistic planning based on epistemic and aleatoric uncertainty estimates in non-stationary environments.

- [5] Ava Pettet, Yunuo Zhang, Baiting Luo, Kyle Wray, Hendrik Baier, Aron Laszka, Abhishek Dubey, and Ayan Mukhopadhyay. "Decision Making in Non-Stationary Environments with Policy-Augmented Search". In: *Proceedings of the 23rd International Conference on Autonomous Agents and Multiagent Systems*. AAMAS '24. Auckland, New Zealand: International Foundation for Autonomous Agents and Multiagent Systems, 2024, pp. 2417–2419. ISBN: 9798400704864.
- Summary: Introduces PA-MCTS algorithm that combines stale policy knowledge with online Monte Carlo tree search to handle non-stationary environments where relearning optimal policies is computationally expensive.

Real-World Applications & Benchmarks

- [6] Nathaniel S. Keplinger, Baiting Luo, Iliyas Bektas, Yunuo Zhang, Kyle Hollins Wray, Aron Laszka, Abhishek Dubey, and Ayan Mukhopadhyay. "NS-Gym: Open-Source Simulation Environments and Benchmarks for Non-Stationary Markov Decision Processes". In: *Proceedings of the Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*. 2025. URL: <https://arxiv.org/abs/2501.09646>.
- Summary: Presents first standardized toolkit for non-stationary MDPs integrated with Gymnasium framework, enabling reproducible evaluation of algorithms under changing environmental conditions.
- [7] Rishav Sen, Yunuo Zhang, Fangqi Liu, Jose Paolo Talusan, Ava Pettet, Yoshinori Suzue, Ayan Mukhopadhyay, and Abhishek Dubey. "Online Decision-Making Under Uncertainty for Vehicle-to-Building Systems". In: *Proceedings of the ACM/IEEE 16th International Conference on Cyber-Physical Systems (ICCPS)*. ICCPS '25. California, USA: Association for Computing Machinery, 2025.
- Summary: Models vehicle-to-building energy optimization as an MDP and uses domain-guided Monte Carlo tree search to handle heterogeneous chargers and demand charges over long planning horizons.
- [8] Alvaro Velasquez, Houbing Song, Pradeep Ravikumar, S. Shankar Sastry, and Sandeep Neema, eds. Wiley-IEEE Press, 2025, p. 512. ISBN: 978-1-394-30237-6. URL: <https://www.wiley.com/en-us/Neuro-symbolic+AI%5C%3A+Foundations+and+Applications-p-9781394302376>.
- Summary: Contributed Chapter 4 on neuro-symbolic approaches for UAV navigation and search planning in urban rescue operations.
- [9] Yunuo Zhang, Baiting Luo, Ayan Mukhopadhyay, Daniel Stojcsics, Daniel Elenius, Anirban Roy, Susmit Jha, Miklos Maroti, Xenofon Koutsoukos, Gabor Karsai, and Abhishek Dubey. "Shrinking POMCP: A Framework for Real-Time UAV Search and Rescue". In: *2024 International Conference on Assured Autonomy (ICAA)*. 2024, pp. 48–57. DOI: [10.1109/ICAA64256.2024.00016](https://doi.org/10.1109/ICAA64256.2024.00016).
- Summary: Presents Shrinking POMCP that reduces computational complexity in UAV search and rescue by dynamically guiding exploration toward non-sparse belief regions.

Awards and Honors

- 2025 **NeurIPS 2025 Scholar Award**
- 2020–2021 **Dean's List (3 semesters)**, *Vanderbilt University*, Fall 2020, Spring 2021, Fall 2021

Teaching and Service

- Professional Service
- 2026 **Reviewer**, *NeurIPS 2026*
The Fortieth Annual Conference on Neural Information Processing Systems
- 2023 **Student Volunteer**, *SMARTCOMP 2023*
IEEE International Conference on Smart Computing
- Teaching Experience
- 2022 Fall **Teaching Assistant**, *Database Management Systems*, Vanderbilt University
- 2021 Fall **Teaching Assistant**, *Algorithms*, Vanderbilt University
- 2020 Fall **Teaching Assistant**, *Discrete Structures*, Vanderbilt University

Core Coursework

Reinforcement Learning, Statistical Foundations of Deep Learning, Foundations of Machine Learning, Advanced Artificial Intelligence, Algorithms for Decision-Making, Automated Verification, Foundations of Hybrid and Embedded Systems

Skills

Programming Languages: Python, C++, Julia, LaTeX

Frameworks & Libraries: AirSim, Isaac Lab, OpenAI Gymnasium, Numpy, Pytorch, Tensorflow